4 Compounds **P** and **Q** are structural isomers.



Fig. 4.1

(a) **(i)** Define structural isomerism.

.....

..... [2]

(b) Fig. 4.3 shows a synthesis starting from **T**, a different isomer of **R** and **S**.

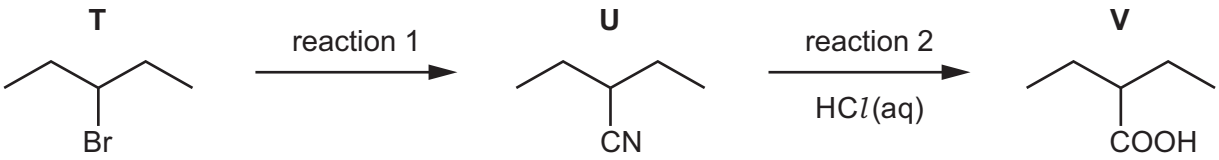


Fig. 4.3

(i) Identify the reagent and conditions for reaction 1.

..... [1]

(ii) Reaction 2 is a hydrolysis reaction.

Construct an equation for reaction 2.

$(C_2H_5)_2CHCN$ ..... [1]

4 The skeletal formula of **E** is shown in Fig. 4.1.

E

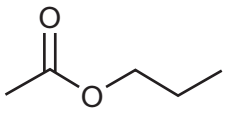


Fig. 4.1

(a) State the molecular formula of **E**.

..... [1]

(d) **F** is an isomer of **E**.
F reacts with HC*l*(aq) to produce **G** and **H**.
G is a secondary alcohol.
H is also produced when HC*l*(aq) is added to CH_3CN.

(i) Construct an equation for the acid hydrolysis of CH_3CN with HC*l*(aq).

..... [1]

6 **(a)** $HOCl$ reacts with $CH_2{=}CH_2$ to produce $HOCH_2CH_2Cl$ in an electrophilic addition reaction.

(i) Define addition reaction.

.....

..... [1]

(c) $HOCH_2CH_2Cl$ reacts in a two-step synthesis to produce $HOCH_2CH_2COOH$, as shown in Fig. 6.2.

$$HOCH_2CH_2Cl \xrightarrow{\text{step 1}} HOCH_2CH_2CN \xrightarrow[HCl(aq)]{\text{step 2}} HOCH_2CH_2COOH$$

Fig. 6.2

(i) State the reagent and conditions for step 1.

..... [1]

(ii) Identify the type of reaction that occurs in step 2.

..... [1]

(iii) Complete the equation to show the reaction in step 2.

$HOCH_2CH_2CN$ + ... [1]

4 **(a)** Propanoic acid, CH_3CH_2COOH, reacts with reducing agent **Q** to produce propan-1-ol, $CH_3CH_2CH_2OH$.

(i) Suggest the formula of reducing agent **Q**.

..... [1]

(b) Propan-1-ol is converted to compound **T** in a three-step synthesis.

In step 1, $SOCl_2$ is added to propan-1-ol to produce compound **E**.

In step 2, **E** reacts with a suitable reagent to produce butanenitrile.

In step 3, butanenitrile is heated with NaOH(aq).

(i) Name the type of reaction that occurs in step 1.

..... [1]

(ii) Identify the reagent and conditions required in step 2.

..... [1]

(iii) Construct an equation to describe the reaction of butanenitrile with NaOH(aq) in step 3.

..... [2]

(iv) Name the type of reaction that occurs in step 3.

..... [1]

4 Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

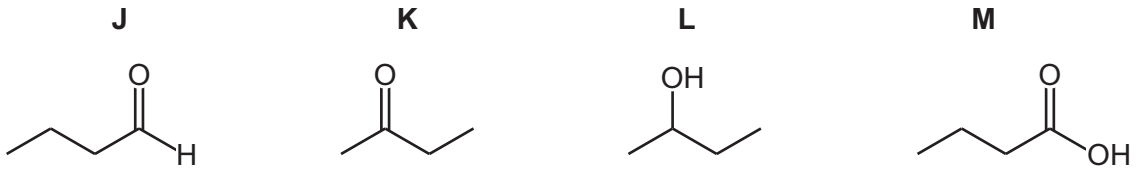


Fig. 4.1

(a) Table 4.1 gives details of tests on **J** to **M**. In each test, only **two** compounds give a positive result.

Complete Table 4.1.

Table 4.1

reagent	observation of positive result	compounds that give a positive result
acidified $K_2Cr_2O_7$(aq)		**J** and **L**
alkaline I_2(aq)	yellow precipitate	
	orange precipitate	**J** and **K**
Na(s)		

[5]

(b) **K** reacts with HCN in the presence of a KCN catalyst, forming **N**.

(i) Complete Fig. 4.2 to show the mechanism for this reaction.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.2

[4]

(ii) Identify the class of compound of which **N** is a member.

..... [1]

4 Propanone, CH_3COCH_3, is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

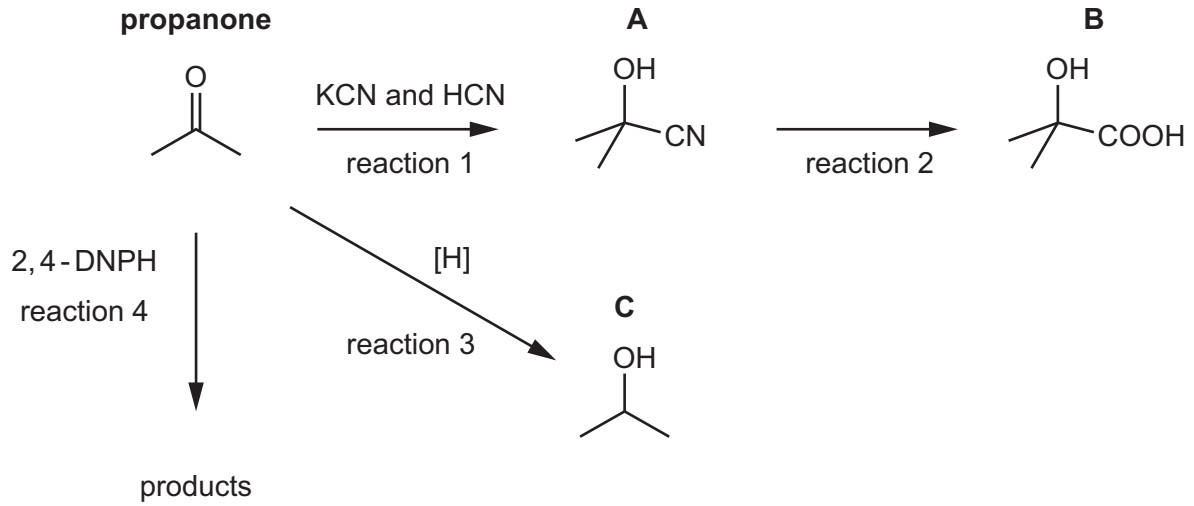


Fig. 4.1

(a) Reaction 1 is a nucleophilic addition reaction.

(i) Complete Fig. 4.2 to show the mechanism for the formation of **A** from propanone.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.2

[3]

(b) Suggest the reagents and conditions for reaction 2.

..... [1]

2 Oxygen is a Group 16 element.

(a) **(i)** Write equations for the following reactions.

- sodium and oxygen

.....

- sulfur and oxygen

.....

[2]

(ii) Write an equation for the reaction of CH_3CN with H_2O in acidic conditions.

CH_3CN + ...H_2O + ...H^+ .....

[1]

4 Propanone, CH_3COCH_3, is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

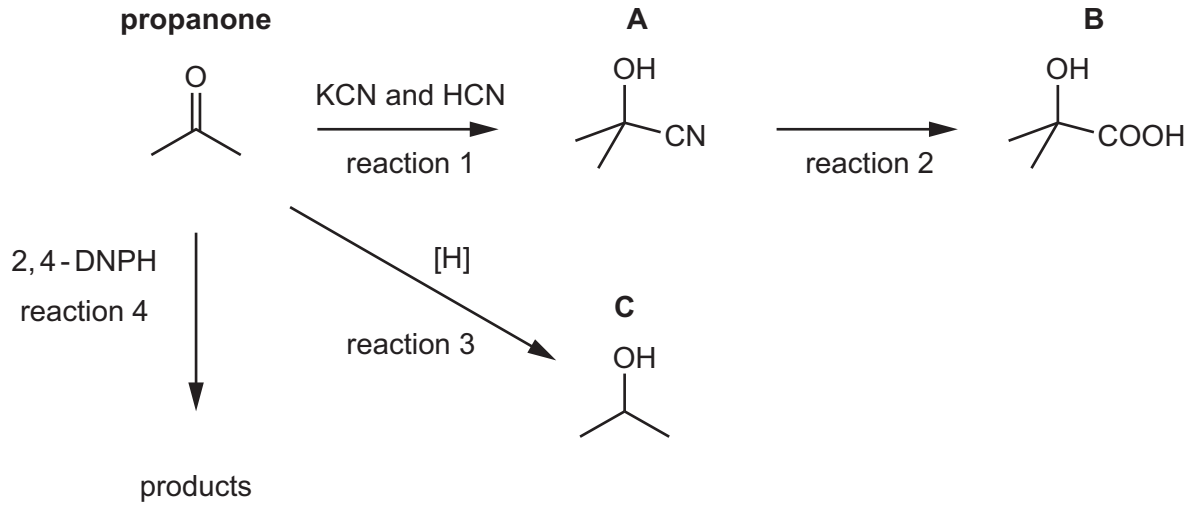


Fig. 4.1

(a) Reaction 1 is a nucleophilic addition reaction.

(i) Complete Fig. 4.2 to show the mechanism for the formation of **A** from propanone.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.2

[3]

(b) Suggest the reagents and conditions for reaction 2.

..... [1]

5 **(a)** **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

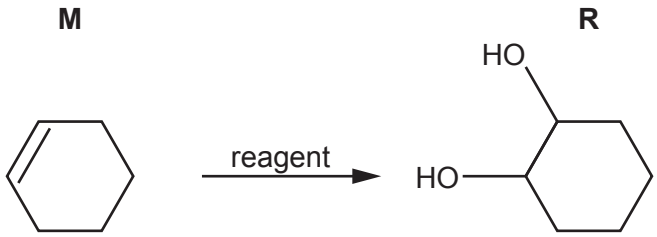


Fig. 5.1

Identify the reagent and conditions for this reaction.

..... [1]

(d) **Y** is made from **Q** in a three-step reaction.

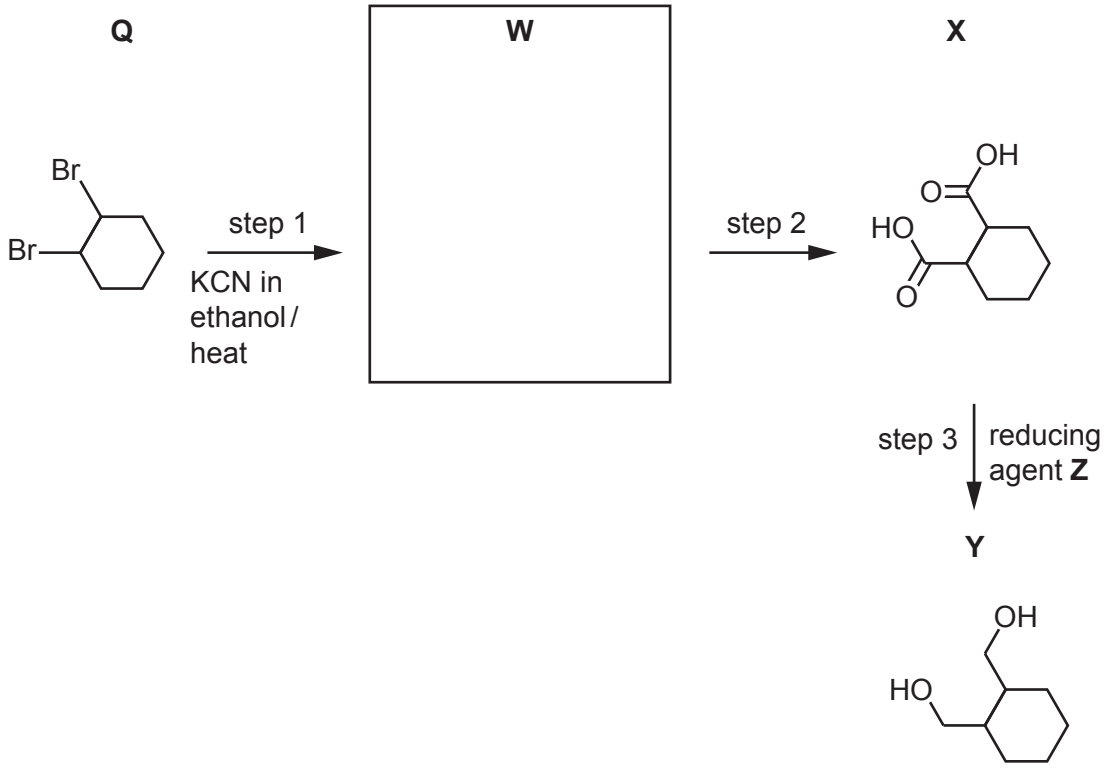


Fig. 5.4

(i) Draw the structure of **W** in the box in Fig. 5.4. [1]

(ii) In step 2, **W** is heated with HCl(aq) to produce **X** and an inorganic product.

Identify the formula of the inorganic product.

..... [1]

4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

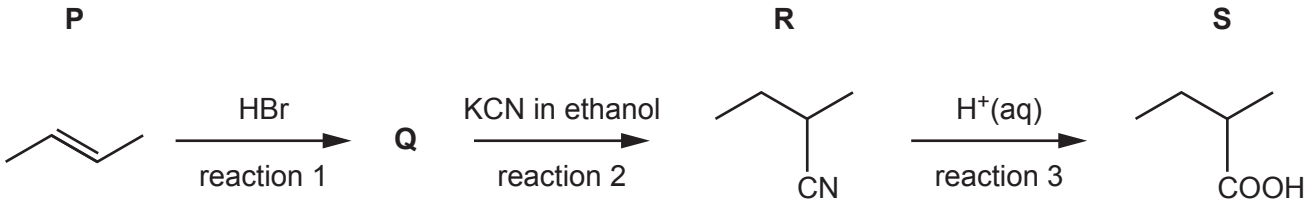


Fig. 4.1

(a) **P**, **Q**, **R** and **S** show stereoisomerism.

Complete Table 4.1 by identifying with a tick (✓) the type of stereoisomerism that each molecule shows.

The type of stereoisomerism shown by **Q** is given.

Table 4.1

	P	**Q**	**R**	**S**
geometrical isomerism				
optical isomerism		✓		

[2]

(iii) Complete the equation for reaction 3. **R** is represented as C_4H_9CN.

C_4H_9CN + ... [1]

2 Barium hydroxide, $Ba(OH)_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $Ba(OH)_2$.

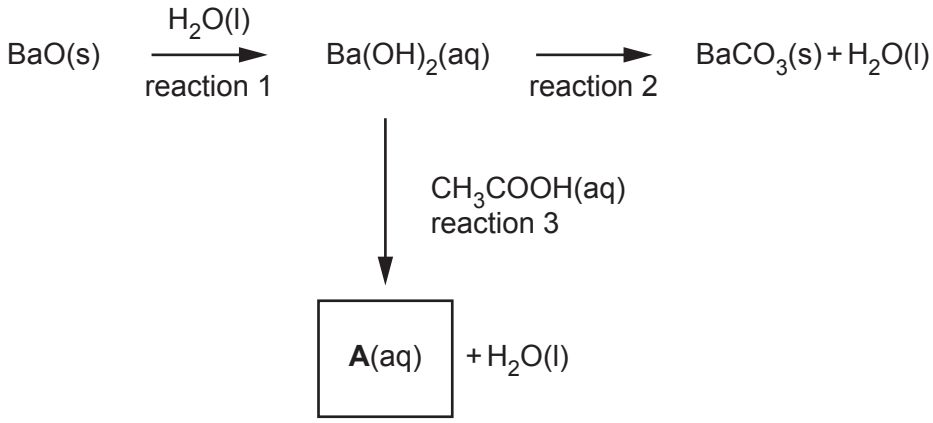


Fig. 2.1

(a) **(i)** State the variation in solubilities of group 2 hydroxides.

..... [1]

(c) $Ba(OH)_2$ is used to hydrolyse organic compounds.

Fig. 2.2 shows the reaction of **B** with $Ba(OH)_2$, followed by acidification.

Draw the structures of the organic products of the process shown in Fig. 2.2.

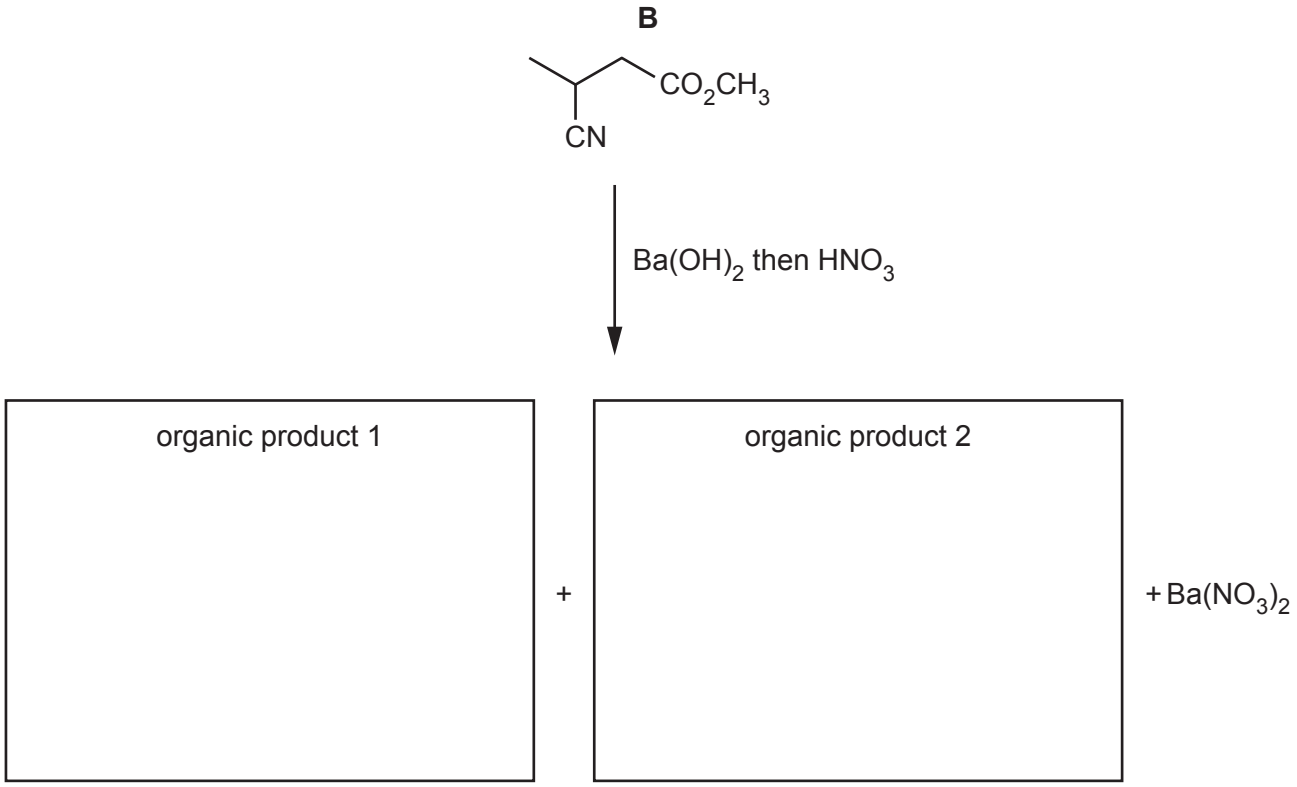


Fig. 2.2

[3]

4 Lactic acid, $CH_3CH(OH)COOH$, and pyruvic acid, $CH_3COCOOH$, both contain two functional groups.

lactic acid **pyruvic acid**

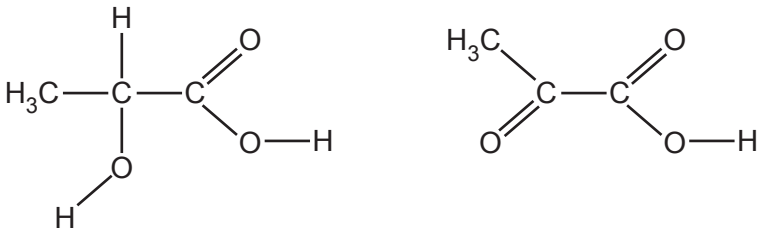


Fig. 4.1

(a) (i) Explain why lactic acid exists as optical isomers.

.....

.....

..... [1]

(b) Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

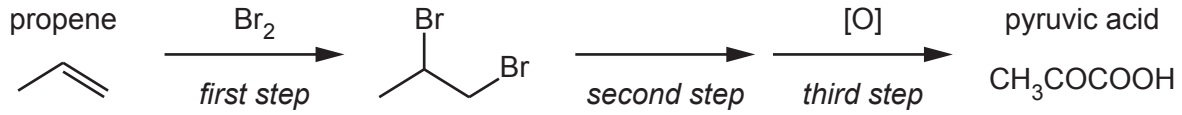


Fig. 4.3

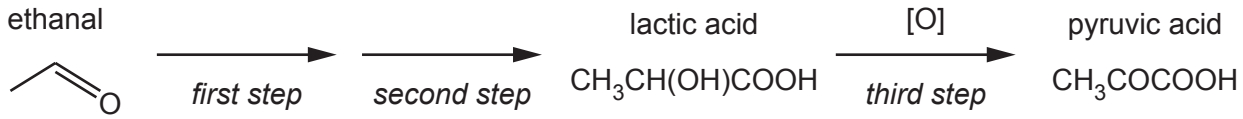


Fig. 4.4

(i) Complete the diagram in Fig. 4.5 to show the mechanism for the reaction of propene with Br_2.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.5

[3]

(iii) Complete Table 4.1 to give details of the reagents and conditions used in each of the two syntheses shown in Fig. 4.3 and Fig. 4.4.

Table 4.1

		synthesis from propene (shown in Fig. 4.3)	synthesis from ethanal (shown in Fig. 4.4)
reagents and conditions used	*first step*	Br_2	
	second step		
	third step		

[4]

2 Barium hydroxide, $Ba(OH)_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $Ba(OH)_2$.

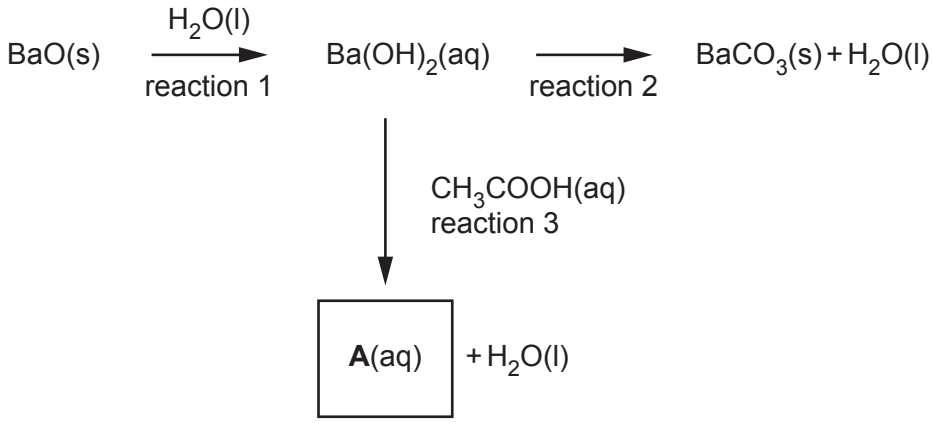


Fig. 2.1

(a) **(i)** State the variation in solubilities of group 2 hydroxides.

..... [1]

(c) $Ba(OH)_2$ is used to hydrolyse organic compounds.

Fig. 2.2 shows the reaction of **B** with $Ba(OH)_2$, followed by acidification.

Draw the structures of the organic products of the process shown in Fig. 2.2.

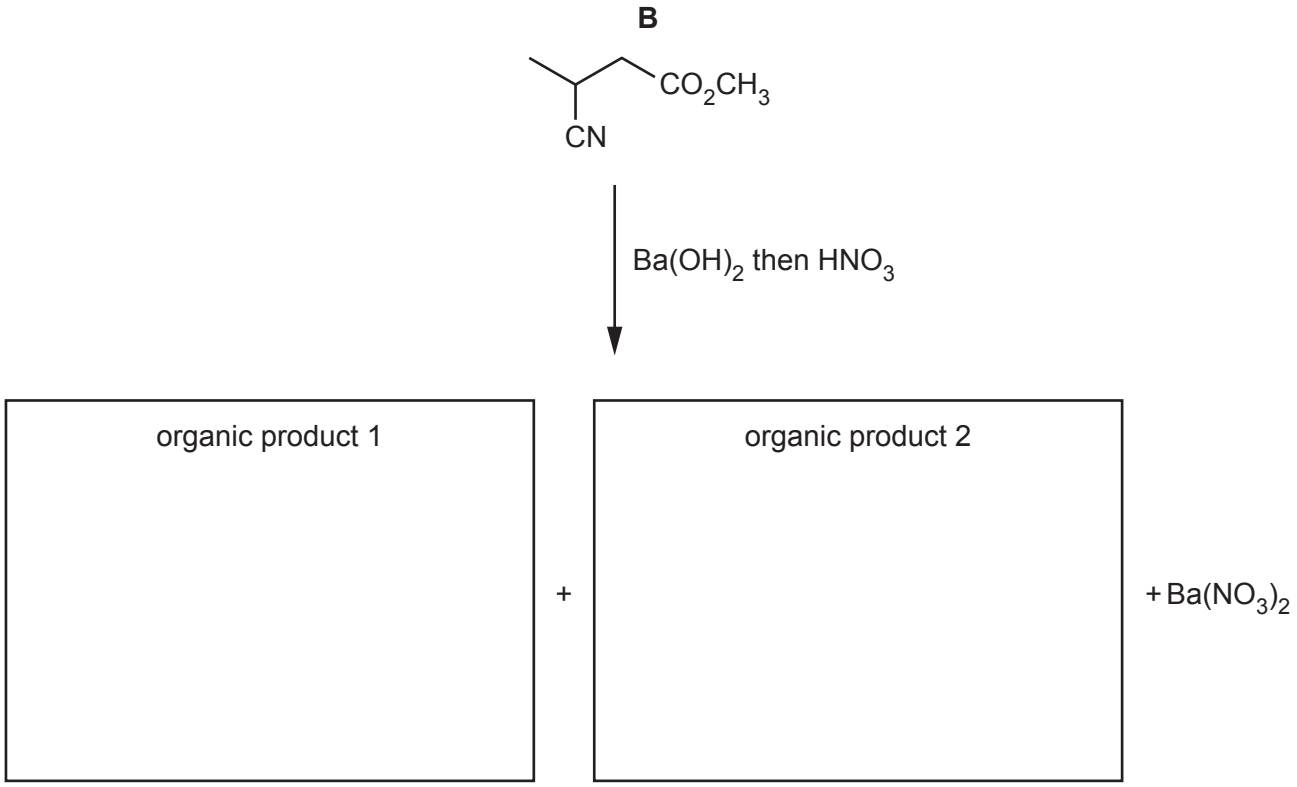


Fig. 2.2

[3]

6 Compound **W**, CH_2=CHCN, is used to make an addition polymer which is present in carbon fibres.

(a) Draw **one** repeat unit of the addition polymer of **W**.

[1]

(f) Propylamine can also be formed in a two-step synthesis from propan-1-ol, as shown in Fig. 6.4.

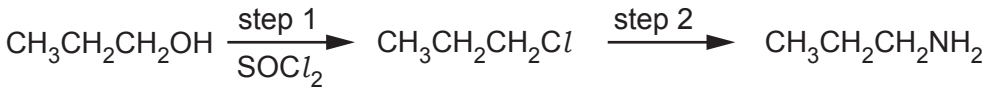


Fig. 6.4

(i) Name the type of reaction in step 1 in Fig. 6.4.

..... [1]

(ii) Identify the reagent and conditions for step 2 in Fig. 6.4.

.....

..... [2]

3 Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from CH_3CH_2Cl.

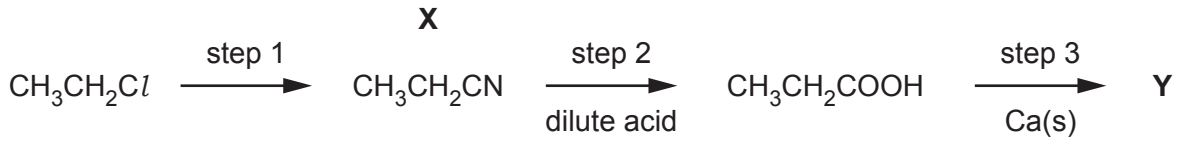


Fig. 3.1

(a) (i) State the reagent and conditions for step 1 in Fig. 3.1.

..... [1]

(iii) Identify the type of reaction that occurs when dilute acid is added to **X** in step 2.

..... [1]